Materials Science
M.Sc. education program

crei.skoltech.ru/cest/education/materials-science-program

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Where to get information

Where to look:

- crei.skoltech.ru/cest/education/materials-science-program

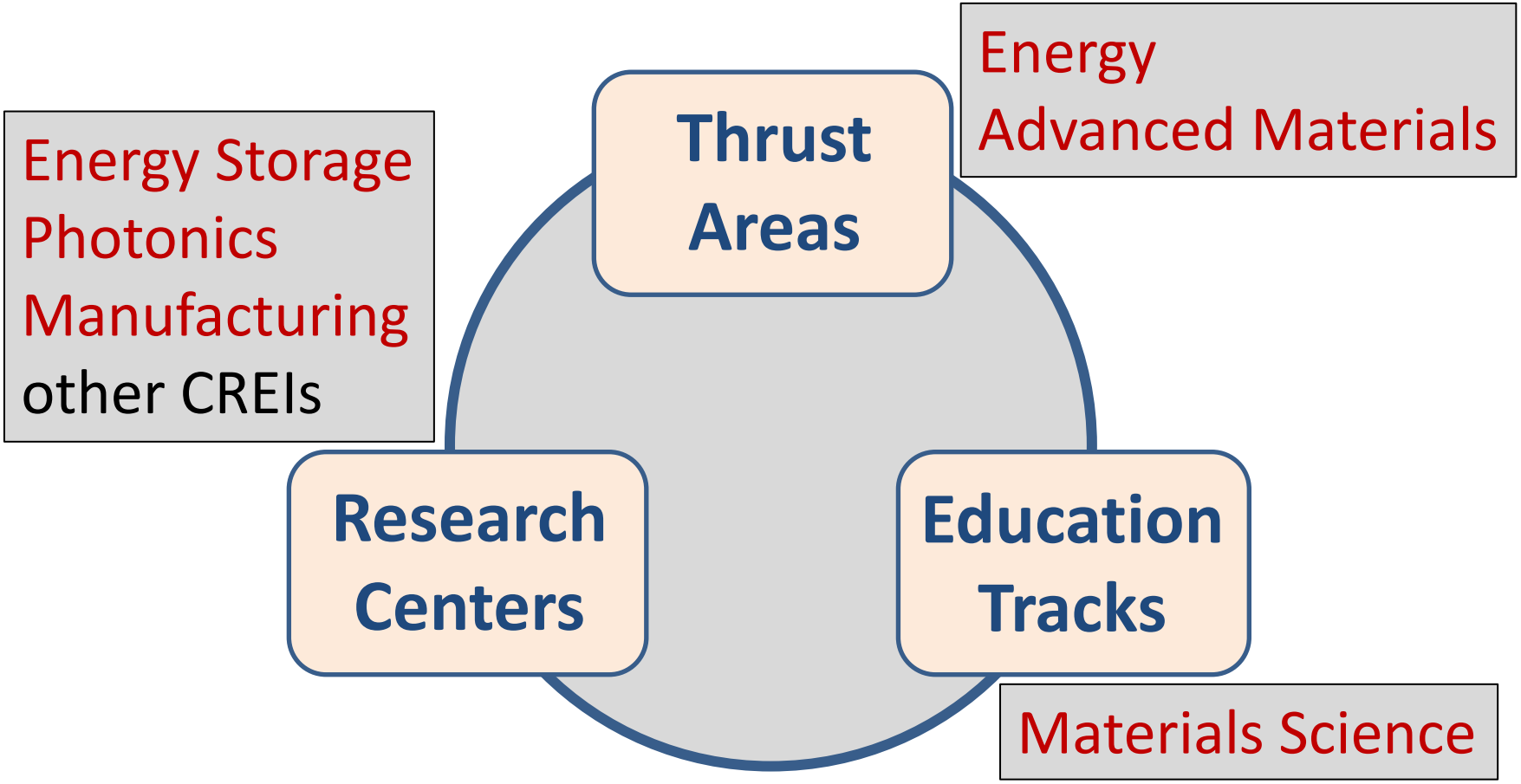
Whom to ask:

- Your academic/research advisor
- Program Coordinator – Andriy Zhugayevych
  - Experimental Materials Science – Artem Abakumov
  - Computational Materials Science – Andriy Zhugayevych
- Education Office

For other contacts see the program web-page (above)

Skoltech research & education environment

- No separate theory or experiment departments
- No separate physics or chemistry departments



Materials Science track: courses

Core courses

- Survey of Materials **(Term 1B)**
- Materials Chemistry **(Term 2)**

Experimental Materials Science

Recommended courses

- Introduction to Solid State Physics**(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Carbon Nanomaterials **(3)**
- Electrochemistry **(4)**
- Materials Selection in Design **(5)**

Computational Materials Science

Recommended courses

- Computational Chemistry and Materials Modeling **(2)**
- Structure & Property of Materials **(3)**
- Advanced Materials Modeling **(4)**
- Materials Selection in Design **(5)**
- Advanced Solid State Physics **(6)**

Elective courses

- discuss with your advisor

Materials Science track: other activities

Understand your load:

- 1 full course = 20 hours per week (except for Term 1B)
- *Hint*: combine research and education

Seminars (1 hour per week):

- Research Seminar "Advanced Materials Science" – core
- Energy Colloquium – optional
- Computational Materials Science seminar – for subtrack
- Your group seminars – per advisor

Background courses (half-course per term):

- Math for Engineers, Quantum Mechanics (Term 1B!)
- English, Writing, Presentation, Pedagogy

Introductory courses from other programs (see webpage)

Advanced courses (see webpage)

M.Sc. Research in Materials Science

In brief

- *Highly interdisciplinary*
- Experimental Materials Science with most publications on:
  - energy storage – Abakumov, Antipov, Stevenson
  - optoelectronics including solar cells – Nasibulin, Troshin
  - materials engineering – Korsunsky
- Computational Materials Science with most publications on:
  - methods – Gonze, Levchenko, Shapeev
  - applications – Oganov, Tretiak, Zhugayevych
  - theoretical chemistry and physics – Buchachenko, Fine
- *… with details on faculty web-pages: example*
- Research topics will be covered on Survey of Materials course

M.Sc. Thesis 2016-2019

- Materials for energy conversion and storage (22 students)
- Materials for optoelectronics (4 students)
- Other materials and computational methods (7 students)

Academic calendar and important deadlines

Academic Year - MSc YEAR 1

advisor, program, II plans

academic mobility contest

27-Aug-18	03-Sep-18	10-Sep-18	17-Sep-18	24-Sep-18	01-Oct-18	08-Oct-18	15-Oct-18	22-Oct-18	29-Oct-18	05-Nov-18	12-Nov-18	19-Nov-18	26-Nov-18	03-Dec-18	10-Dec-18	17-Dec-18	04-Feb-19	11-Feb-19	18-Feb-19	25-Feb-19	04-Mar-19	11-Mar-19	06-May-19	13-May-19	20-May-19	27-May-19	03-Jun-19	10-Jun-19	17-Jun-19	24-Jun-19	01-Jul-19	08-Jul-19	15-Jul-19	22-Jul-19	29-Jul-19	05-Aug-19	12-Aug-19	19-Aug-19
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Core courses

Education, get basic skills

Practice

Academic Year - MSc YEAR 2

project

draft

final

27-Aug-18	03-Sep-18	10-Sep-18	17-Sep-18	24-Sep-18	01-Oct-18	08-Oct-18	15-Oct-18	22-Oct-18	29-Oct-18	05-Nov-18	12-Nov-18	19-Nov-18	26-Nov-18	03-Dec-18	10-Dec-18	17-Dec-18	24-Dec-18	31-Dec-18	07-Jan-19	14-Jan-19	21-Jan-19	28-Jan-19	04-Feb-19	11-Feb-19	18-Feb-19	25-Feb-19	04-Mar-19	11-Mar-19	18-Mar-19	25-Mar-19	01-Apr-19	08-Apr-19	15-Apr-19	22-Apr-19	29-Apr-19	06-May-19	13-May-19	20-May-19	27-May-19	03-Jun-19	10-Jun-19	17-Jun-19	24-Jun-19	01-Jul-19	08-Jul-19	15-Jul-19	22-Jul-19	29-Jul-19	05-Aug-19	12-Aug-19	19-Aug-19
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V	C	C	C	C	C	C	E	E	C	C	C	C	C	C	E	E	V	V	C	C	C	C	C	C	C	C	C	E	E	C	C	C	C	C	V	V	C	E	E	MS	MS	MS	MS	V	V	V	V	V	V	V	V

Thesis research

Defense

KEY:

- Innovation Workshop
- Credit-bearing activity (course, research)
- Evaluation period (assessment and application period)
- Industrial Immersion
- Independent Studies Period
- Vacation
- MSc Thesis Defence